

■ SCRATCH SIMPLE ANIMATION

Grade 2 · Coding Lesson

Make your sprite walk, dance, and talk — with code!

■ Learning Objectives

- Understand what coding and animation are
- Navigate the Scratch interface
- Use Motion, Looks, Sound, and Control blocks
- Build a complete animated Scratch project

■ What is Coding?

Coding is **telling the computer what to do!** Just like giving step-by-step instructions to a friend, we give instructions to the computer — and it follows every step exactly.

■ What is Animation?

Animation is **making pictures move!** You see animations every day — in cartoons, movies, and games. In Scratch we build our own animations using colourful code blocks.

The Scratch Interface

Scratch has **4 main parts** that work together to help you create animations. Knowing where everything lives makes coding much easier!

■ Stage

The big white area where your animation plays. Press the green flag ■ to start!

■ Sprite Pane

Shows all your characters (sprites). Click a sprite to select it and add code to it.

■ Block Palette

The colourful blocks on the left — Motion (blue), Looks (purple), Sound (pink), Control (orange) and more!

■ Scripts Area

Drag blocks here and snap them together to build your code. This is your coding workspace!

Blocks for Animation

■ Motion Blocks

Move 10 steps
Turn & rotate
Glide to position

■ Looks Blocks

Next costume
Say 'Hello!'
Change size

■ Control Blocks

Repeat 10
Forever
Wait 1 second

STEP 1 Choose Your Sprite!

A **sprite** is the character or object you want to animate in Scratch. Click the sprite library icon (bottom right of the sprite pane) and choose your favourite!

1. Animals ■
2. People ■
3. Fantasy ■
4. Objects ■

■ Teacher Tip

Tip: There are hundreds of sprites — you can even draw your own!

STEP 2 Make Your Sprite Move!

Use the '**move 10 steps**' block to make your sprite walk across the screen. Find it in the Motion blocks (blue). Drag it into the scripts area. Click the block — your sprite takes 10 steps forward!

1. Go to Block Palette → Motion (blue)
2. Drag 'move 10 steps' to scripts area
3. Stack it under 'When Green Flag clicked'
4. Press ■ and watch it go!

■ Teacher Tip

Try: Change the number from 10 to 50 — what happens?

STEP 3 Costume Change!

A **costume** is a different look for your sprite. Switching costumes quickly creates the illusion of movement — just like a cartoon flip-book!

1. Click on 'Looks' blocks (purple)
2. Drag 'next costume' into your script
3. Place it after 'move 10 steps'
4. Press ■ — see it animate!

■ Teacher Tip

Tip: Check the Costumes tab to see all available looks for your sprite.

STEP 4 Add Loops!

A **loop** repeats your code again and again — no need to stack the same blocks dozens of times. The **'repeat 10'** block is found in Control (orange).

1. Go to Control blocks (orange)
2. Drag 'repeat 10' into your script
3. Place 'move 10 steps' and 'next costume' INSIDE the loop
4. Press ■ — smooth animation!

■ Teacher Tip

Try: Use 'forever' instead of 'repeat 10' for non-stop animation!

STEP 5 Add Sound!

Make your animation even more fun with sound! The **'play sound Meow'** block is in Sound blocks (pink). Add it inside your loop for a sound on every step.

1. Go to Sound blocks (pink)
2. Drag 'play sound Meow' into your script
3. Put it inside your repeat loop
4. Press ■ and listen ■

■ Teacher Tip

Explore: Open the Sounds tab to record your own voice!

See It All Together! ■

Here is a complete Scratch animation script. Each block snaps onto the next:

■	When Green Flag clicked	START — runs when ■ is pressed
■	repeat 10	LOOP — repeat everything inside 10 times
■	move 10 steps	MOTION — sprite walks forward
■	next costume	LOOKS — sprite changes appearance



play sound Meow

SOUND — sprite makes a sound



end repeat

END of the loop

■ YOUR TURN TO ANIMATE!

It's YOUR time to create! Follow the checklist below, then experiment and make your animation totally unique. **There is no wrong answer — have fun!**

-  Choose a sprite from the library
-  Add a "When Green Flag clicked" block
-  Add a "repeat 10" loop block (orange)
-  Put "move 10 steps" inside the loop
-  Add "next costume" inside the loop
-  Add "play sound Meow" inside the loop
-  Press the green flag and see your animation! ■

■ Stretch Challenge

Can you add a BACKGROUND to your animation? Click the Stage, choose a backdrop from the library, and make your sprite look like it is walking through a forest, city, or outer space!

What Did We Learn? ■

Coding

Coding means giving the computer step-by-step instructions. Scratch uses colourful blocks instead of typed words!

Scratch

Scratch is a free, block-based coding tool. It has a Stage, Sprites, Block Palette, and Scripts Area.

Animation

Animation is making pictures move. Switching costumes quickly creates the effect of movement.

Loops

Loops (like 'repeat 10') run code multiple times without stacking the same blocks over and over.

■ Keep exploring Scratch and make your own stories! — scratch.mit.edu